

Custom Carrier Board Design for LattePanda Mu Compute Module

Project Overview

Designed a custom 4-layer carrier board for the LattePanda Mu x86 compute module using KiCad 10, based on the open-source reference schematics and hardware design files provided by DFRobot. The objective was to develop a compact carrier platform exposing key high-speed and peripheral interfaces while applying industry-standard PCB design practices for signal integrity and manufacturability.

Design Scope

The project encompassed schematic capture, component selection, PCB layout, stack-up planning, and generation of manufacturing outputs. Major interfaces incorporated into the design include:

- USB 2.0
- USB 3.0 SuperSpeed
- HDMI video output
- Gigabit Ethernet
- Power input and distribution circuitry
- Expansion and peripheral connectivity

The design required interpretation of reference documentation and integration of multiple high-speed interfaces within a constrained PCB footprint.

High-Speed Design Considerations

A primary focus of the project was applying high-speed PCB design principles during layout development.

Key considerations included:

- Differential pair routing for USB 3.0 and Ethernet interfaces
- Length matching of high-speed differential signals
- Continuous return current paths through reference planes
- Minimization of routing discontinuities and signal stubs
- Strategic placement of vias and layer transitions
- Ground plane integrity and signal isolation

Controlled impedance requirements were evaluated using the NextPCB impedance calculator to determine appropriate trace geometries based on the selected 4-layer stack-up.

Power Distribution and Layout Strategy

The PCB layout was developed with attention to power and signal integrity principles, including:

- Dedicated power and ground plane utilization
- Decoupling capacitor placement near power pins
- Power distribution routing optimization
- Functional partitioning of high-speed and support circuitry
- Connector placement to minimize routing complexity

The 4-layer stack-up was selected to provide solid reference planes and improve routing efficiency for high-speed interfaces.

Layer Distribution

- 1) Top: Signal + GND pours
- 2) Inner layer 1 : GND Pour
- 3) Inner layer 2 : POWER (3.3V, 1V, 15V,12V)
- 4) Bottom : Signal + GND pours

Tools and Technologies

- KiCad 10 (schematic capture and PCB layout)
- NextPCB online Impedance Calculator
- DFRobot LattePanda Mu Open-Source Hardware Reference Design

Skills:

- Multi-layer PCB design
- High-speed digital layout practices
- Differential pair routing
- Controlled impedance planning
- USB 3.0, HDMI, and Ethernet interface integration
- Schematic capture and hardware documentation interpretation
- Design for Manufacturing (DFM)
- Signal Integrity (SI) aware PCB layout

This project provided practical experience in designing a carrier board for a modern x86 compute module and strengthened understanding of high-speed PCB layout techniques, impedance-controlled routing, and multi-layer board development workflows.